Roll No. | | | | | | | | | | |

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem.-4)

PHYSICAL PHARMACEUTICS-II

Subject Code : BP-403T

M.Code : 75845

Date of Examination: 05-05-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. Answer briefly:

a) What is meant by projected diameter of a particle?

b) What is the porosity during cubic packing?

c) Mention the angle of repose indicating poor, moderate and excellent flow?

d) What is specific surface area?

e) What is dilatant flow?

f) Give two examples of photolytic degradation of drugs.

g) What is pseudo zero order reaction?

h) What is yield value and in which types of systems it is observed?

i) What is specific surface area and its importance?

j) Write the equation for calculating the half life for a drug following first order degradation kinetics.

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SECTION - B

2. Give a detailed account of the electrical properties of colloids. Explain the significance of Nernst and Zeta potential in influencing stability of colloidal systems.

3. Explain plastic and elastic deformation in solids and correlate with Heckel Plot.

4. Explain the use of accelerated stability testing of pharmaceutical products using temperature as a stress condition.

SECTION - C

5. What are emulsions? Discuss the role of primary and auxiliary emulsifiers in formulating stable emulsions. Briefly write about the tests employed for evaluating stability of emulsions.

6. Explain coacervation of colloids.

7. Write a note on thixotropy.

8. Explain the role of structured vehicle while formulating flocculated suspensions.

9. Give a brief account of different methods for particle size measurement (principle only).

10. Write a note on prevention of hydrolytic degradation in pharmaceutical products.

11. Explain particle shapes and their influence on packing.

12. Briefly explain the differences between conventional and microemulsions.

13. Enumerate the methods used for determination of surface area of powders. Discuss any one method in detail.

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NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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